

F24-008-D-IntraSec

Project Team

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Session 2021-2025

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June, 2025

Contents

1	Introduction	1
1.1	Problem Statement	1
1.2	Scope	1
1.3	Modules	2
1.3.1	Module 1: Assets Inventory	2
1.3.2	Module 2: Risk Assessment	2
1.3.3	Module 3: Report Generation	3
1.3.4	Module 4: Patch Deployment	3
1.3.5	Module 5: Alerts and Notifications	3
1.4	User Classes and Characteristics	4
2	Project Requirements	5
2.1	Use-case Diagram	5
2.2	High-Level Use Case Table	6
2.3	Expanded Use Cases	7
2.3.1	UC-01: Discover/Manage Assets	7
2.3.2	UC-02 : Deploy Patches	10
2.3.3	UC-03 : Search Vulnerabilities	13
2.3.4	UC-04 : Manage Users	16
2.3.5	UC-05 : Receive Alerts Notification	19
2.3.6	UC-06 : Manage Reports	22
2.3.7	UC-07 :Initiate Risk Assessment	25
2.4	Functional Requirements	28
2.4.1	Module 1: Assets Inventory	28
2.4.2	Module 2: Risk Assessment	28
2.4.3	Module 3: Report Generation	28
2.4.4	Module 4: Patch Deployment	29
2.4.5	Module 5: Alerts and Notifications	29
2.5	Non-Functional Requirements	30
2.5.1	Reliability	30
2.5.2	Usability	30
2.5.3	Performance	30

2.5.4	Security	30
3	System Overview	33
3.0.1	Functionality (Modules)	33
3.1	Architectural Design	35
3.1.1	Architecture Diagram	35
3.1.2	Deployment Diagram	36
3.2	Design Models	37
3.2.1	Data-flow Diagrams	37
3.2.2	Activity Diagrams	38
3.2.2.1	Discover Assets	38
3.2.2.2	Assess Risks	39
3.2.2.3	Generate Report	40
3.2.2.4	Deploy Patches	41
3.2.3	System-level Sequence Diagrams	42
3.2.3.1	Discover Assets	42
3.2.3.2	Manage Assets	43
3.2.3.3	Deploy Patches	44
3.2.3.4	Assess Risks	45
3.3	Data Design	46
3.3.1	Data Structures	46
3.3.1.1	Key Entities	46
3.3.2	Elasticsearch Indexes	47
3.3.3	PostgreSQL Databases:	48
	References	49

List of Figures

2.1	Use-Case Diagram	5
3.1	Architecture Diagram	35
3.2	Deployment Diagram	36
3.3	Data-flow Diagram - Level 0	37
3.4	Data-flow Diagram - Level 2	37
3.5	Discover Assets - Activity Diagram	38
3.6	Assess Risk - Activity Diagram	39
3.7	Generate Report - Activity Diagram	40
3.8	Deploy Patches - Activity Diagram	41
3.9	Discover Asset - System Sequence Diagram	42
3.10	Manage Assets - System Sequence Diagram	43
3.11	Deploy Patches - System Sequence Diagram	44
3.12	Assess Risks - System Sequence Diagram	45
3.13	CVE JSON Schema	47
3.14	CPE JSON Schema	47

List of Tables

1.1	User Classes and Their Characteristics	4
2.1	High-Level Use Case Table for IntraSec with System Administrator . . .	6
2.2	Expanded Use Case: Discover/Manage Assets	7
2.3	Expanded Use Case: Deploy Patches	10
2.4	Expanded Use Case: Search Vulnerabilities	13
2.5	Expanded Use Case: Manage Users	16
2.6	Expanded Use Case: Receive Alerts and Notifications	19
2.7	Expanded Use Case: Manage Reports	22
2.8	Expanded Use Case: Initiate Risk Assessment	25

Chapter 1

Introduction

IntraSec is a client-less risk assessment and asset management tool , with the goal of providing real-time inventory of network assets and identifying security risks across the internal network without requiring agent/client installation individual machines.

1.1 Problem Statement

In large organizations, as the network constantly expands and becomes more and more complex. It becomes harder to keep track of the *assets*, *devices*, and *software* being used. Organizations are at risk of security gaps and struggle to meet industry standards without a centralized system for tracking assets and assessing risks.

Furthermore, the manual methods of assessing risks, prioritizing them, and deploying patches are not just time-consuming but also susceptible to mistakes.

IntraSec seeks to address these issues by providing a centralized, automated, and client-free solution that can effectively monitor assets, conduct real-time risk evaluations and deploy patches to improve security and assets management.

1.2 Scope

The scope of **IntraSec** encompasses the development and deployment of a client-less risk assessment and asset management system specifically designed for **internal networks**.

It includes discovering and inventorying all devices, services, and software within the network. The system continuously tracks these assets in real-time, ensuring up-to-date information. *IntraSec* integrates with the *NIST* vulnerability database to automatically assess risks associated with network assets and assigns a risk score based on the severity

of detected vulnerabilities.

It generates clear, actionable reports that help network administrators prioritize risk mitigation efforts. One of the key functionalities is the automatic deployment of patches to address vulnerabilities. In cases where automatic patching isn't available, administrators are alerted for manual intervention.

Additionally, the system provides continuous monitoring, real-time alerts, and notifications for critical security issues.

1.3 Modules

1.3.1 Module 1: Assets Inventory

The Assets Inventory module is responsible for discovering and tracking all devices, software, and services within the internal network. It ensures up-to-date information on network assets by continuously monitoring changes in real-time.

1. **Front-end:** User interface for viewing and managing asset details.
2. **Back-end:** Active Directory integration for centralized asset management.
3. **Back-end:** Real-time discovery of hosts, software, and services.
4. **Back-end:** Continuous tracking and updating of asset details in *Postgresql*

1.3.2 Module 2: Risk Assessment

The Risk Assessment module analyzes the assets identified in the inventory, assessing them for potential vulnerabilities using the *NIST* vulnerability database and *ElasticSearch*. It assigns a risk score based on severity, helping to prioritize remediation efforts.

1. **Front-end:** Dashboard for displaying risk assessment results and scores.
2. **Back-end:** Integration with the *NIST* Vulnerability Database.
3. **Back-end:** Vulnerability scanning using *ElasticSearch*.
4. **Back-end:** Assignment of risk scores to each asset based on vulnerability severity.
5. **Back-end:** Prioritization of remediation efforts for high-risk assets.

1.3.3 Module 3: Report Generation

After risk assessment, *IntraSec* generates detailed, easy-to-understand reports. These reports provide an overview of the network's security posture, including identified risks and recommended mitigation steps.

1. **Front-end:** Interface for viewing and exporting generated reports.
2. **Back-end:** Standardized risk scoring for consistency in evaluations.
3. **Back-end:** Detailed analysis of detected vulnerabilities and their impact.
4. **Back-end:** Recommendations for remediation of critical threats.
5. **Back-end:** Exportable reports for documentation and audits.

1.3.4 Module 4: Patch Deployment

The Patch Deployment module handles the automated application of patches to fix vulnerabilities identified during the risk assessment. It ensures that the network stays protected against known threats.

1. **Front-end:** Interface for monitoring patch deployment status.
2. **Back-end:** Automated patch deployment to vulnerable assets.
3. **Back-end:** Monitoring and reporting of patch deployment progress.
4. **Back-end:** Notifications for manual intervention when automatic patching is unavailable.

1.3.5 Module 5: Alerts and Notifications

This module provides real-time alerts and notifications regarding security issues or vulnerabilities that require immediate attention. It ensures that administrators are aware of critical threats as they arise.

1. **Front-end:** User interface for viewing alerts and notifications.
2. **Back-end:** Real-time alerts for vulnerabilities and security incidents.
3. **Back-end:** Customizable thresholds for alert generation.
4. **Back-end:** Notifications for vulnerabilities requiring urgent attention.

1.4 User Classes and Characteristics

Table 1.1: User Classes and Their Characteristics

User Class	Description
Network Security Team	This team is responsible for monitoring the internal networks. They will use IntraSec to assess vulnerabilities and take proactive measures to secure the network. Their focus is on threat detection and mitigation.
IT Administrators	They oversee the installation and upkeep of IntraSec. Their role includes configuring the system and ensuring it integrates smoothly with existing network management tools. They are crucial for the day-to-day operation of the system.
System Administrators	These users manage the technical side of the network, including patch deployments. They will rely on IntraSec to automate patch management and improve efficiency in addressing vulnerabilities.

Chapter 2

Project Requirements

2.1 Use-case Diagram

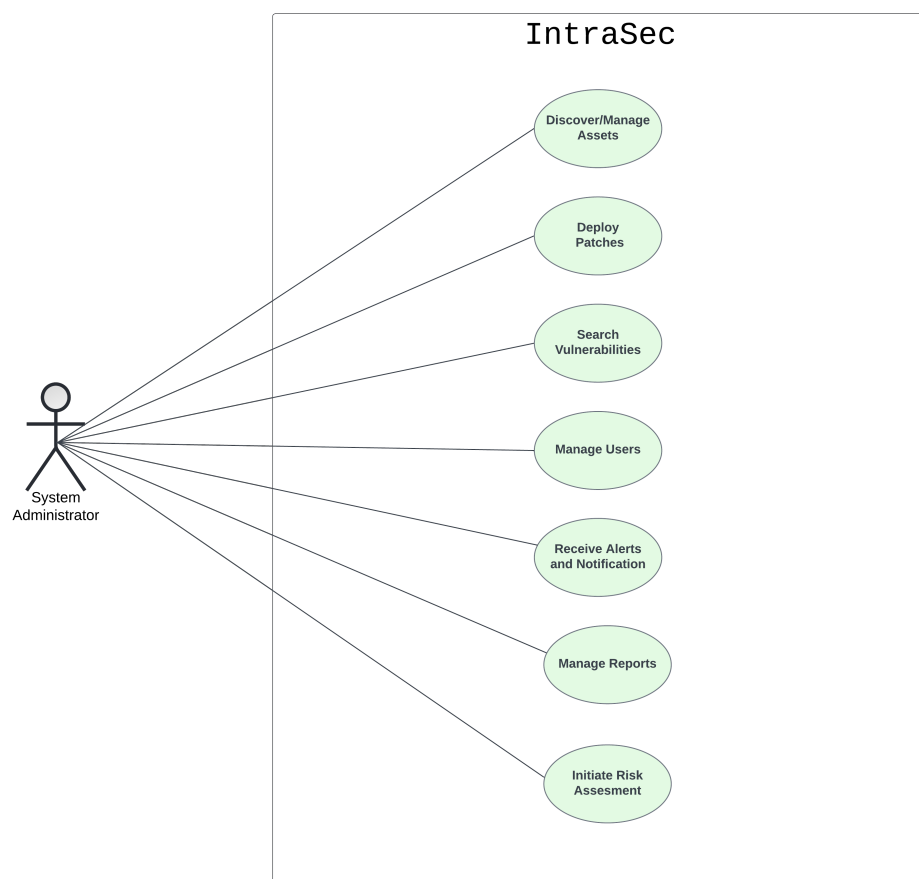


Figure 2.1: Use-Case Diagram

2.2 High-Level Use Case Table

Use Case	Actor	Type	Description
Discover/Manage Assets	System Administrator	Primary	Discover all assets on the network and manage the asset inventory, ensuring an up-to-date record of devices.
Deploy Patches	System Administrator	Primary	Deploy patches across the network to address identified vulnerabilities and ensure asset security.
Search Vulnerabilities	System Administrator	Supporting	Search the vulnerability database for specific assets, types of vulnerabilities, or other search parameters.
Manage Users	System Administrator	Supporting	Manage user roles, permissions, and access to the IntraSec system.
Receive Alerts/Notifications	System Administrator	Supporting	Receive alerts and notifications about critical vulnerabilities or abnormal activities requiring immediate attention.
Manage Reports	System Administrator	Primary	Generate, view, and manage risk and vulnerability reports for system review and decision-making.
Initiate Risk Assessment	System Administrator	Primary	Conduct a risk assessment by analyzing network assets for potential vulnerabilities based on predefined criteria.

Table 2.1: High-Level Use Case Table for IntraSec with System Administrator

2.3 Expanded Use Cases

2.3.1 UC-01: Discover/Manage Assets

Table 2.2: Expanded Use Case: Discover/Manage Assets

Attribute	Details
Use Case ID	UC-01
Use Case Name	Discover/Manage Assets
Actor	System Administrator
Type	Primary
Description	This use case allows the System Administrator to discover all assets connected to the network, maintain an up-to-date inventory, and manage asset details effectively. It involves scanning the network for devices, identifying their types, statuses, and configurations, and making necessary updates to the asset database.
Preconditions	• The System Administrator has access to the IntraSec application. • The network is operational and the administrator has the necessary permissions to scan for assets.
Postconditions	• The asset inventory is updated with the latest discovered devices. • The asset management system reflects any changes made during the process.

Attribute	Details	
Basic Flow		
	User Action	System Response
	The System Administrator logs into the IntraSec application.	The system verifies the credentials and grants access to the application.
	The administrator navigates to the "Asset Management" section.	The system displays the asset management interface with options to discover, view, and manage assets.
	The administrator initiates an asset discovery scan.	The system starts scanning the network for connected devices.
	The administrator waits for the scan to complete.	The system identifies all connected assets and compiles their details.
	The system presents the list of discovered assets to the administrator.	The administrator reviews the list, which includes asset types, statuses, and configurations.
	The administrator selects an asset to view its details.	The system retrieves and displays detailed information about the selected asset.
Alternative Flows	• If no assets are found: The system notifies the administrator and prompts to check the network connection or permissions. • If the administrator chooses to cancel the scan: The system terminates the discovery process and returns to the asset management interface.	
Exceptions	• The system encounters network errors during the discovery scan. • The administrator does not have sufficient permissions to access certain asset details.	

Attribute	Details
Frequency of Use	This use case is performed regularly as part of network management tasks, typically initiated weekly or after significant changes to the network.
Special Requirements	• The discovery process should comply with network security protocols. • The system should provide a user-friendly interface for reviewing and managing assets.

2.3.2 UC-02 : Deploy Patches

Table 2.3: Expanded Use Case: Deploy Patches

Attribute	Details
Use Case ID	UC-02
Use Case Name	Deploy Patches
Actor	System Administrator
Type	Primary
Description	This use case allows the System Administrator to deploy patches to identified vulnerabilities across the networked assets. It includes selecting the patches to be applied, verifying the compatibility with existing systems, and monitoring the deployment process.
Preconditions	• The System Administrator has access to the IntraSec application. • Vulnerabilities have been identified and patches are available for deployment.
Postconditions	• The vulnerabilities are addressed with the appropriate patches applied. • The system logs all deployment actions for future reference.

Attribute	Details	
Basic Flow	User Action	System Response
	The System Administrator logs into the IntraSec application.	The system verifies the credentials and grants access to the application.
	The administrator navigates to the "Patch Management" section.	The system displays the list of available patches and their details.
	The administrator selects the patches to deploy.	The system verifies compatibility with the current asset configurations.
	The administrator initiates the patch deployment process.	The system starts deploying the selected patches to the identified assets.
	The administrator monitors the deployment progress.	The system provides real-time updates on the status of the patch deployment.
	The system completes the deployment and provides a summary report.	The administrator reviews the summary report and confirms successful patching.
	Alternative Flows	• If a patch fails to deploy: The system notifies the administrator and provides options to troubleshoot the issue. • If the administrator chooses to cancel the deployment: The system halts the process and reverts any partial changes made.
Exceptions	• The system encounters compatibility issues with specific assets. • The administrator lacks the necessary permissions to deploy certain patches.	
Frequency of Use	This use case is performed regularly as part of system maintenance, typically initiated after security audits or vulnerability assessments.	

Attribute	Details
Special Requirements	• The deployment process should comply with established change management protocols. • The system should provide detailed logging for all actions taken during the patch deployment.

2.3.3 UC-03 : Search Vulnerabilities

Table 2.4: Expanded Use Case: Search Vulnerabilities

Attribute	Details
Use Case ID	UC-03
Use Case Name	Search Vulnerabilities
Actor	System Administrator
Type	Primary
Description	This use case allows the System Administrator to search for known vulnerabilities in the asset inventory based on various criteria such as severity, asset type, and status. It aids in identifying and prioritizing vulnerabilities for remediation.
Preconditions	• The System Administrator has access to the IntraSec application. • The asset inventory is populated with assets and associated vulnerabilities.
Postconditions	• A list of vulnerabilities matching the search criteria is displayed for review. • The system logs the search query for auditing purposes.

Attribute	Details	
Basic Flow		
	User Action	System Response
	The System Administrator logs into the IntraSec application.	The system verifies the credentials and grants access to the application.
	The administrator navigates to the "Vulnerability Management" section.	The system displays the search interface for vulnerabilities.
	The administrator specifies search criteria (e.g., severity, asset type).	The system processes the search parameters and prepares to query the database.
	The administrator initiates the search for vulnerabilities.	The system retrieves and displays a list of vulnerabilities matching the search criteria.
	The administrator reviews the list of vulnerabilities.	The system presents detailed information about each vulnerability, including severity and affected assets.
Alternative Flows	• If no vulnerabilities are found: The system notifies the administrator and suggests refining the search criteria. • If the administrator chooses to cancel the search: The system stops the search process and returns to the vulnerability management interface.	
Exceptions	• The system encounters issues accessing the vulnerability database. • The administrator lacks the necessary permissions to view certain vulnerabilities.	
Frequency of Use	This use case is performed regularly as part of security assessments, typically conducted after new vulnerabilities are disclosed or during scheduled audits.	

Attribute	Details
Special Requirements	• The search process should comply with data privacy and security policies. • The system should provide an intuitive interface for specifying search criteria and reviewing results.

2.3.4 UC-04 : Manage Users

Table 2.5: Expanded Use Case: Manage Users

Attribute	Details
Use Case ID	UC-04
Use Case Name	Manage Users
Actor	System Administrator
Type	Primary
Description	This use case allows the System Administrator to manage user accounts, including adding, modifying, or deleting user information, assigning roles and permissions, and ensuring secure access control within the IntraSec system.
Preconditions	• The System Administrator has the necessary permissions to manage user accounts. • The system is operational and user data is accessible.
Postconditions	• User accounts are updated according to the actions performed by the administrator. • Role-based access control is correctly applied to user accounts.

Attribute	Details														
Basic Flow	<table border="1"> <thead> <tr> <th data-bbox="502 286 874 331">User Action</th><th data-bbox="874 286 1340 331">System Response</th></tr> </thead> <tbody> <tr> <td data-bbox="502 331 874 465">The System Administrator logs into the IntraSec application.</td><td data-bbox="874 331 1340 465">The system verifies the credentials and grants access to the application.</td></tr> <tr> <td data-bbox="502 465 874 600">The administrator navigates to the "User Management" section.</td><td data-bbox="874 465 1340 600">The system displays the current list of user accounts.</td></tr> <tr> <td data-bbox="502 600 874 734">The administrator selects the option to add, modify, or delete a user.</td><td data-bbox="874 600 1340 734">The system provides the necessary input forms for the selected action.</td></tr> <tr> <td data-bbox="502 734 874 869">The administrator enters or updates user details (e.g., username, role).</td><td data-bbox="874 734 1340 869">The system validates the input and saves the user data.</td></tr> <tr> <td data-bbox="502 869 874 1003">The administrator assigns roles and permissions to the user.</td><td data-bbox="874 869 1340 1003">The system applies role-based access control and updates the user account accordingly.</td></tr> <tr> <td data-bbox="502 1003 874 1182">The system confirms the completion of the user management task and displays a summary.</td><td data-bbox="874 1003 1340 1182">The administrator reviews and confirms the changes made to the user account.</td></tr> </tbody> </table>	User Action	System Response	The System Administrator logs into the IntraSec application.	The system verifies the credentials and grants access to the application.	The administrator navigates to the "User Management" section.	The system displays the current list of user accounts.	The administrator selects the option to add, modify, or delete a user.	The system provides the necessary input forms for the selected action.	The administrator enters or updates user details (e.g., username, role).	The system validates the input and saves the user data.	The administrator assigns roles and permissions to the user.	The system applies role-based access control and updates the user account accordingly.	The system confirms the completion of the user management task and displays a summary.	The administrator reviews and confirms the changes made to the user account.
User Action	System Response														
The System Administrator logs into the IntraSec application.	The system verifies the credentials and grants access to the application.														
The administrator navigates to the "User Management" section.	The system displays the current list of user accounts.														
The administrator selects the option to add, modify, or delete a user.	The system provides the necessary input forms for the selected action.														
The administrator enters or updates user details (e.g., username, role).	The system validates the input and saves the user data.														
The administrator assigns roles and permissions to the user.	The system applies role-based access control and updates the user account accordingly.														
The system confirms the completion of the user management task and displays a summary.	The administrator reviews and confirms the changes made to the user account.														
Alternative Flows	• If the administrator attempts to delete a user with active sessions: The system prompts the administrator to terminate the sessions before proceeding. • If the administrator assigns invalid roles or permissions: The system notifies the administrator and suggests valid options.														
Exceptions	• The administrator lacks permissions to modify certain user accounts. • The system encounters data integrity issues during the update process.														

Attribute	Details
Frequency of Use	This use case is performed as needed when user accounts require updates, or when new users are added to the system.
Special Requirements	• All changes to user accounts must be logged for auditing purposes. • The system should enforce strong password policies during account creation.

2.3.5 UC-05 : Receive Alerts Notification

Table 2.6: Expanded Use Case: Receive Alerts and Notifications

Attribute	Details
Use Case ID	UC-05
Use Case Name	Receive Alerts and Notifications
Actor	System Administrator
Type	Primary
Description	This use case allows the System Administrator to receive real-time alerts and notifications about critical vulnerabilities, security threats, or system updates. It includes configurable alert settings, immediate notifications for critical issues, and a dashboard view of all current alerts.
Preconditions	• The System Administrator has the necessary access to the IntraSec application. • Alerts and notifications are configured in the system based on network monitoring and risk assessment rules.
Postconditions	• The System Administrator is notified about the relevant security events. • The system logs all alerts and notifications for review and audit purposes.

Attribute	Details														
Basic Flow	<table> <tr> <th>User Action</th><th>System Response</th></tr> <tr> <td>The System Administrator logs into the IntraSec application.</td><td>The system verifies credentials and provides access to the dashboard.</td></tr> <tr> <td>The administrator configures the alert settings for specific vulnerabilities or security events.</td><td>The system saves the configuration and sets up notifications based on the defined rules.</td></tr> <tr> <td>A security threat or system event occurs that triggers an alert.</td><td>The system generates an alert and sends notifications through the configured channels (email, SMS, dashboard).</td></tr> <tr> <td>The administrator receives the alert in real-time.</td><td>The system displays the alert with details on the dashboard and sends any additional notifications as configured.</td></tr> <tr> <td>The administrator views the alert details and decides on further actions.</td><td>The system provides detailed information about the event, including risk severity and suggested remediation steps.</td></tr> <tr> <td>The administrator marks the alert as acknowledged or resolved.</td><td>The system logs the response and updates the alert status.</td></tr> </table>	User Action	System Response	The System Administrator logs into the IntraSec application.	The system verifies credentials and provides access to the dashboard.	The administrator configures the alert settings for specific vulnerabilities or security events.	The system saves the configuration and sets up notifications based on the defined rules.	A security threat or system event occurs that triggers an alert.	The system generates an alert and sends notifications through the configured channels (email, SMS, dashboard).	The administrator receives the alert in real-time.	The system displays the alert with details on the dashboard and sends any additional notifications as configured.	The administrator views the alert details and decides on further actions.	The system provides detailed information about the event, including risk severity and suggested remediation steps.	The administrator marks the alert as acknowledged or resolved.	The system logs the response and updates the alert status.
User Action	System Response														
The System Administrator logs into the IntraSec application.	The system verifies credentials and provides access to the dashboard.														
The administrator configures the alert settings for specific vulnerabilities or security events.	The system saves the configuration and sets up notifications based on the defined rules.														
A security threat or system event occurs that triggers an alert.	The system generates an alert and sends notifications through the configured channels (email, SMS, dashboard).														
The administrator receives the alert in real-time.	The system displays the alert with details on the dashboard and sends any additional notifications as configured.														
The administrator views the alert details and decides on further actions.	The system provides detailed information about the event, including risk severity and suggested remediation steps.														
The administrator marks the alert as acknowledged or resolved.	The system logs the response and updates the alert status.														
Alternative Flows	• If the system is unable to deliver alerts due to a communication issue: The system retries sending notifications and logs the failure for further investigation. • If the administrator configures incorrect alert thresholds: The system notifies the administrator of the misconfiguration and suggests corrective actions.														

Attribute	Details
Exceptions	• The system encounters connectivity issues while sending notifications. • The administrator lacks the necessary permissions to modify alert settings.
Frequency of Use	This use case is performed continuously as alerts are generated based on real-time monitoring and security events.
Special Requirements	• The system should support configurable notification channels (email, SMS, etc.). • All alerts and notifications must be logged and accessible for audits.

2.3.6 UC-06 : Manage Reports

Table 2.7: Expanded Use Case: Manage Reports

Attribute	Details
Use Case ID	UC-06
Use Case Name	Manage Reports
Actor	System Administrator
Type	Primary
Description	This use case allows the System Administrator to generate, view, and export security reports based on vulnerability assessments and patch deployment statuses. It includes generating detailed reports, analyzing vulnerabilities, and exporting reports for documentation or audits.
Preconditions	• The System Administrator has access to the IntraSec application. • Risk assessments and patch deployments have been performed, and relevant data is available.
Postconditions	• A detailed report is generated with vulnerability analysis, risk scores, and recommended actions. • The report can be exported in various formats (e.g., PDF, CSV) for future reference or audits.

Attribute	Details														
Basic Flow	<table> <tr> <th>User Action</th><th>System Response</th></tr> <tr> <td>The System Administrator logs into the IntraSec application.</td><td>The system verifies credentials and grants access to the dashboard.</td></tr> <tr> <td>The administrator navigates to the "Reports" section.</td><td>The system displays the list of available reports and options to generate new reports.</td></tr> <tr> <td>The administrator selects the type of report (e.g., vulnerability report, patch deployment report).</td><td>The system gathers relevant data based on the selected report type and prepares it for generation.</td></tr> <tr> <td>The administrator configures report parameters (e.g., date range, asset selection).</td><td>The system applies the selected filters and generates a detailed report.</td></tr> <tr> <td>The system generates the report.</td><td>The administrator views the generated report on the dashboard.</td></tr> <tr> <td>The administrator exports the report in the desired format (e.g., PDF, CSV).</td><td>The system generates the report in the chosen format and provides a download link.</td></tr> </table>	User Action	System Response	The System Administrator logs into the IntraSec application.	The system verifies credentials and grants access to the dashboard.	The administrator navigates to the "Reports" section.	The system displays the list of available reports and options to generate new reports.	The administrator selects the type of report (e.g., vulnerability report, patch deployment report).	The system gathers relevant data based on the selected report type and prepares it for generation.	The administrator configures report parameters (e.g., date range, asset selection).	The system applies the selected filters and generates a detailed report.	The system generates the report.	The administrator views the generated report on the dashboard.	The administrator exports the report in the desired format (e.g., PDF, CSV).	The system generates the report in the chosen format and provides a download link.
User Action	System Response														
The System Administrator logs into the IntraSec application.	The system verifies credentials and grants access to the dashboard.														
The administrator navigates to the "Reports" section.	The system displays the list of available reports and options to generate new reports.														
The administrator selects the type of report (e.g., vulnerability report, patch deployment report).	The system gathers relevant data based on the selected report type and prepares it for generation.														
The administrator configures report parameters (e.g., date range, asset selection).	The system applies the selected filters and generates a detailed report.														
The system generates the report.	The administrator views the generated report on the dashboard.														
The administrator exports the report in the desired format (e.g., PDF, CSV).	The system generates the report in the chosen format and provides a download link.														
Alternative Flows	• If the report fails to generate: The system notifies the administrator and provides error details for troubleshooting. • If the administrator chooses to cancel the report generation: The system halts the process and discards the incomplete report.														
Exceptions	• The system encounters missing or incomplete data for report generation. • The administrator lacks the necessary permissions to access certain reports.														

Attribute	Details
Frequency of Use	This use case is performed regularly after vulnerability assessments or patch deployments. Reports may also be generated during audits or compliance checks.
Special Requirements	• The system should support multiple export formats (e.g., PDF, CSV) for reports. • Detailed logging of report generation and export actions should be available for auditing purposes.

2.3.7 UC-07 :Initiate Risk Assessment

Table 2.8: Expanded Use Case: Initiate Risk Assessment

Attribute	Details
Use Case ID	UC-07
Use Case Name	Initiate Risk Assessment
Actor	System Administrator
Type	Primary
Description	This use case allows the System Administrator to initiate a risk assessment of networked assets. The risk assessment analyzes identified assets, assesses their vulnerabilities using the <i>NIST</i> Vulnerability Database, and assigns risk scores based on the severity of vulnerabilities.
Preconditions	• The System Administrator has access to the IntraSec application. • Assets have been identified and listed in the asset inventory. • The system is connected to the <i>NIST</i> Vulnerability Database (ElasticSearch)
Postconditions	• Vulnerabilities in the networked assets are identified and assigned risk scores. • A detailed risk assessment report is generated, highlighting critical vulnerabilities.

Attribute	Details														
Basic Flow	<table> <tr> <th>User Action</th><th>System Response</th></tr> <tr> <td>The System Administrator logs into the IntraSec application.</td><td>The system verifies credentials and grants access to the dashboard.</td></tr> <tr> <td>The administrator navigates to the "Risk Assessment" section.</td><td>The system displays a list of available assets and the option to initiate a new risk assessment.</td></tr> <tr> <td>The administrator selects the assets to be assessed.</td><td>The system loads the selected assets and their details for vulnerability analysis.</td></tr> <tr> <td>The administrator initiates the risk assessment.</td><td>The system queries the <i>NIST</i> Vulnerability Database (Elastic-Search) and begins scanning the assets for known vulnerabilities.</td></tr> <tr> <td>The system assigns risk scores based on the identified vulnerabilities.</td><td>The administrator reviews the progress as the system calculates risk scores based on vulnerability severity.</td></tr> <tr> <td>The system completes the risk assessment and provides a summary report.</td><td>The administrator views the detailed risk assessment report, which includes vulnerabilities, risk scores, and recommended remediation steps.</td></tr> </table>	User Action	System Response	The System Administrator logs into the IntraSec application.	The system verifies credentials and grants access to the dashboard.	The administrator navigates to the "Risk Assessment" section.	The system displays a list of available assets and the option to initiate a new risk assessment.	The administrator selects the assets to be assessed.	The system loads the selected assets and their details for vulnerability analysis.	The administrator initiates the risk assessment.	The system queries the <i>NIST</i> Vulnerability Database (Elastic-Search) and begins scanning the assets for known vulnerabilities.	The system assigns risk scores based on the identified vulnerabilities.	The administrator reviews the progress as the system calculates risk scores based on vulnerability severity.	The system completes the risk assessment and provides a summary report.	The administrator views the detailed risk assessment report, which includes vulnerabilities, risk scores, and recommended remediation steps.
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The system completes the risk assessment and provides a summary report.	The administrator views the detailed risk assessment report, which includes vulnerabilities, risk scores, and recommended remediation steps.														
Alternative Flows	• If the risk assessment process is interrupted: The system notifies the administrator and provides an option to resume or restart the process. • If the administrator cancels the risk assessment: The system halts the process and discards any partially collected data.														

Attribute	Details
Exceptions	• The system encounters network issues or fails to connect to the <i>NIST</i> Vulnerability Database (ElasticSearch). • The system is unable to assess some assets due to missing or incomplete information.
Frequency of Use	This use case is typically performed after significant changes to the network, during routine security assessments, or following a security audit.
Special Requirements	• The system must maintain a reliable connection to the <i>NIST</i> Vulnerability Database (ElasticSearch). • The system should generate detailed logs of the assessment for future audits or troubleshooting.

2.4 Functional Requirements

2.4.1 Module 1: Assets Inventory

The Assets Inventory module is responsible for discovering and tracking all devices, software, and services within the internal network.

1. The system shall discover and display all devices, software, and services in real-time.
2. The system shall allow users to view, filter, and manage the details of assets through a front-end interface.
3. The system shall continuously track changes in asset details, including new additions or removals, and update them in the PostgreSQL database.
4. The system shall integrate with Active Directory to centralize asset management.

2.4.2 Module 2: Risk Assessment

The Risk Assessment module assesses assets for potential vulnerabilities and assigns a risk score to prioritize remediation efforts.

1. The system shall perform vulnerability scanning of assets using *ElasticSearch*.
2. The system shall continuously sync with the NIST Vulnerability Database API to ensure the most up-to-date vulnerability information.
3. The system shall calculate and assign risk scores based on the severity of detected vulnerabilities.
4. The system shall provide a dashboard to display risk scores, vulnerability details, and recommendations for remediation.

2.4.3 Module 3: Report Generation

The Report Generation module provides detailed reports on the network's security posture, risk assessment results, and recommended mitigation steps.

1. The system shall generate standardized reports that include identified vulnerabilities, their severity, and the corresponding risk score.

2. The system shall provide exportable reports for auditing and documentation purposes.
3. The system shall generate detailed analysis reports with recommendations for remediation of critical threats.
4. The system shall allow users to view, filter, and export reports through a front-end interface.

2.4.4 Module 4: Patch Deployment

The Patch Deployment module automates the application of patches to fix vulnerabilities and protect the network from known threats.

1. The system shall automate the deployment of patches for identified vulnerabilities.
2. The system shall provide users with an interface to monitor the status of patch deployment.
3. The system shall notify users if manual intervention is required when automatic patch deployment fails.
4. The system shall report the progress and status of patch deployment.

2.4.5 Module 5: Alerts and Notifications

The Alerts and Notifications module ensures that administrators are informed in real-time of any critical security issues or vulnerabilities that require attention.

1. The system shall generate real-time alerts for detected vulnerabilities or security incidents.
2. The system shall allow users to customize alert thresholds for specific types of vulnerabilities or incidents.
3. The system shall provide an interface for users to view active alerts and notifications.
4. The system shall send notifications for vulnerabilities that require immediate action.

2.5 Non-Functional Requirements

This section outlines the quantifiable non-functional requirements that specify the system's quality attributes, ensuring reliability, usability, performance, and security.

2.5.1 Reliability

- The system shall aim to achieve an up-time of at least **95%** during operational hours, with scheduled maintenance windows allowed.
- In case of a system failure, the system shall log the event and notify administrators within **5 minutes** of detection.
- The system shall recover from critical failures and resume operation within **10 minutes** of the failure being detected.

2.5.2 Usability

- Users shall be able to perform key tasks such as viewing asset information and generating reports in **3 interactions or fewer**.
- **80%** of new users shall be able to navigate and complete common tasks without needing external documentation after **10 minutes** of interaction with the system.

2.5.3 Performance

- The system, based on preliminary estimates and assumptions about network size and performance, shall be able to generate an asset inventory report for up to **100 assets** within approximately **15 minutes**.
- The dashboard page shall fully load within **5 seconds** for **80%** of users on a **>10 Mbps** internet connection.
- Database queries shall return results within **5 seconds** for typical operations, such as retrieving asset details or vulnerability data.

2.5.4 Security

- **80%** of administrative users shall be required to use multi-factor authentication (MFA) for logging into the system, if implemented.

- The system shall maintain logs of access to sensitive data for at least **3 months**, accessible to authorized administrators only.
- The system shall lock user accounts after **5 consecutive failed login attempts**, with a notification sent to the administrator within **10 minutes**.

Chapter 3

System Overview

IntraSec is designed to provide a comprehensive solution for internal network risk assessment and asset management. Its primary objective is to facilitate the discovery, management, and remediation of vulnerabilities across various assets within an active directory domain network.

3.0.1 Functionality (Modules)

The system encompasses several key functionalities and each functionality is handled by its respective micro-services , which communicate via RESTful APIs. The main functionalities are listed below:

- ***Asset Discovery and Management:*** The system allows system administrators to discover all assets connected to the network, maintain an up-to-date inventory, and manage asset details effectively. This includes scanning the network for devices, identifying their types, statuses, and configurations, and making necessary updates to the asset database.
- ***Risk Assessment:*** IntraSec integrates with the Common Vulnerabilities and Exposures (CVE) and Common Platform Enumerations (CPE) databases to identify and assess vulnerabilities associated with the discovered assets. This assessment helps in recognizing potential security threats and prioritizing them for remediation.
- ***Patch Management:*** The system supports the deployment of patches to address identified vulnerabilities. It maintains a repository of patches associated with specific CVEs, facilitating timely updates to mitigate risks.
- ***Reporting:*** IntraSec generates detailed reports summarizing the vulnerabilities identified, the assets affected, and the recommended actions. This reporting functionality ensures that compliance with security protocols and standards is maintained.

This system overview provides a high-level understanding of IntraSec's objectives and functionality. The subsequent sections will delve deeper into the specific use cases, data structures, and implementation details of the system.

3.1 Architectural Design

3.1.1 Architecture Diagram

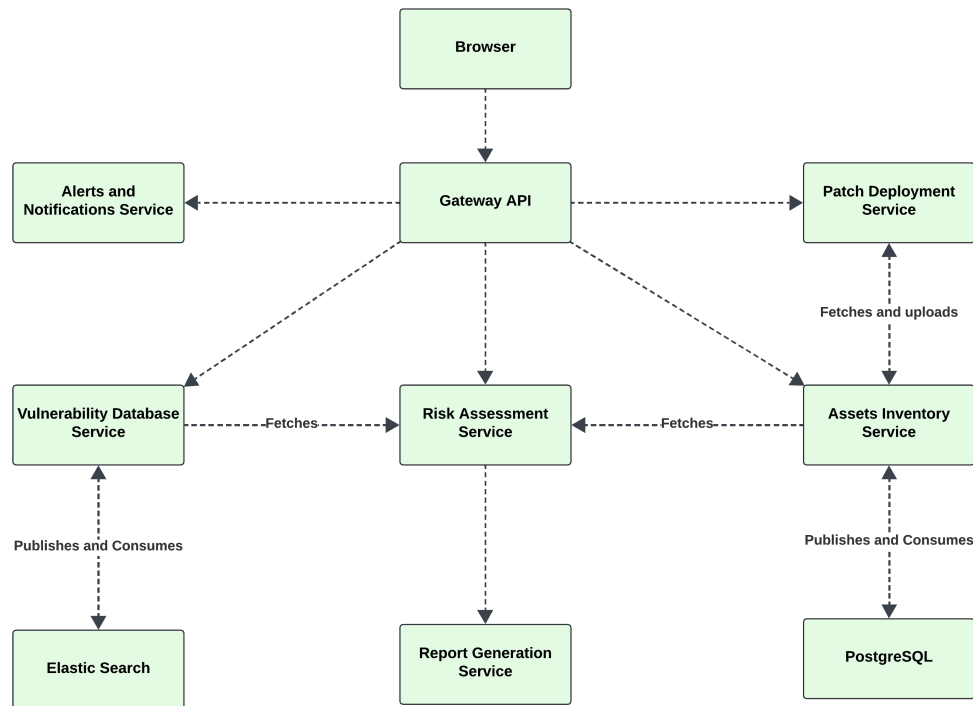


Figure 3.1: Architecture Diagram

The architecture design of **IntraSec** is based on Micro-services and their relations with each other.

Browser is the client-side which communicates with the back-end represented by **Gateway API**. **Gateway API** then communicates with all services based on the input of the admin. Furthermore, Micro-services communicate with each other based on the required data.

For example, **Risk Assessment Service** requires data from **Assets Inventory** and **Vulnerability database** to perform the Risk Analysis. After the risk analysis, **Risk Assessment Service** commands the report generation service to generate easy to understand reports. On the other hand, **Patch Deployment** needs to communicate with the assets inventory service to fetch info and upload the possible patches.

3.1.2 Deployment Diagram

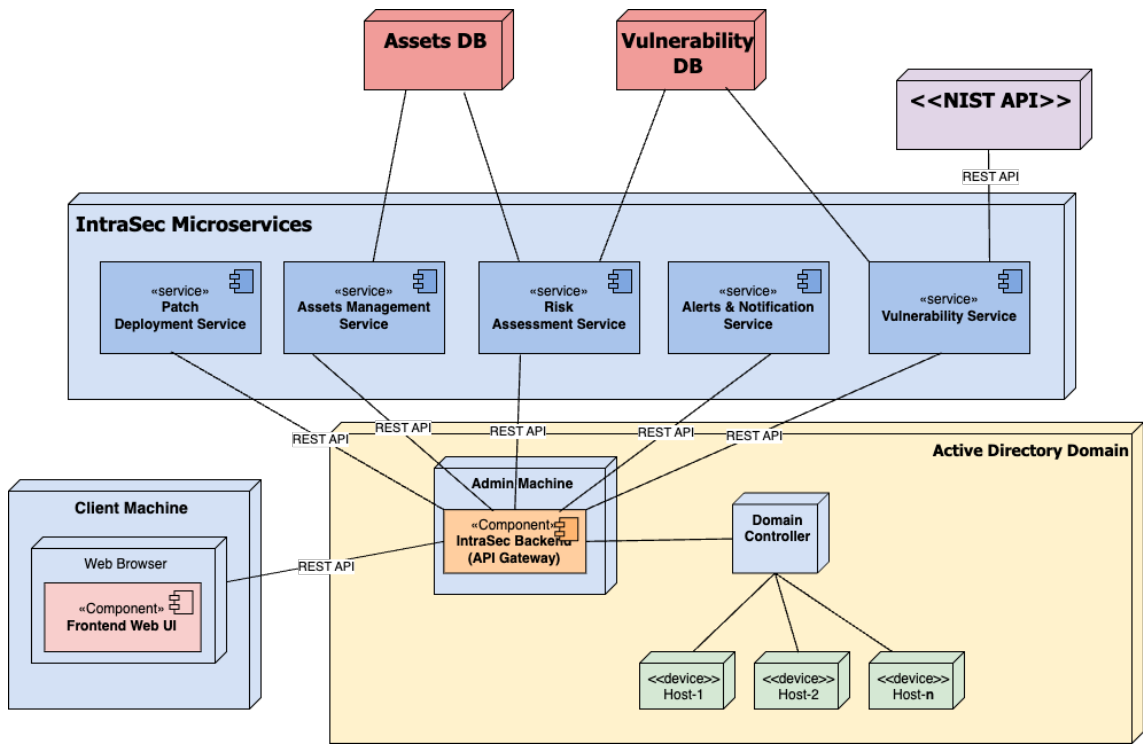


Figure 3.2: Deployment Diagram

3.2 Design Models

3.2.1 Data-flow Diagrams

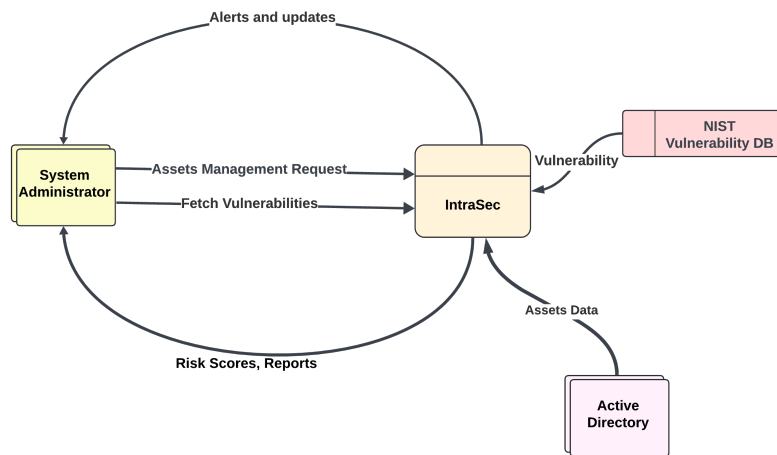


Figure 3.3: Data-flow Diagram - Level 0

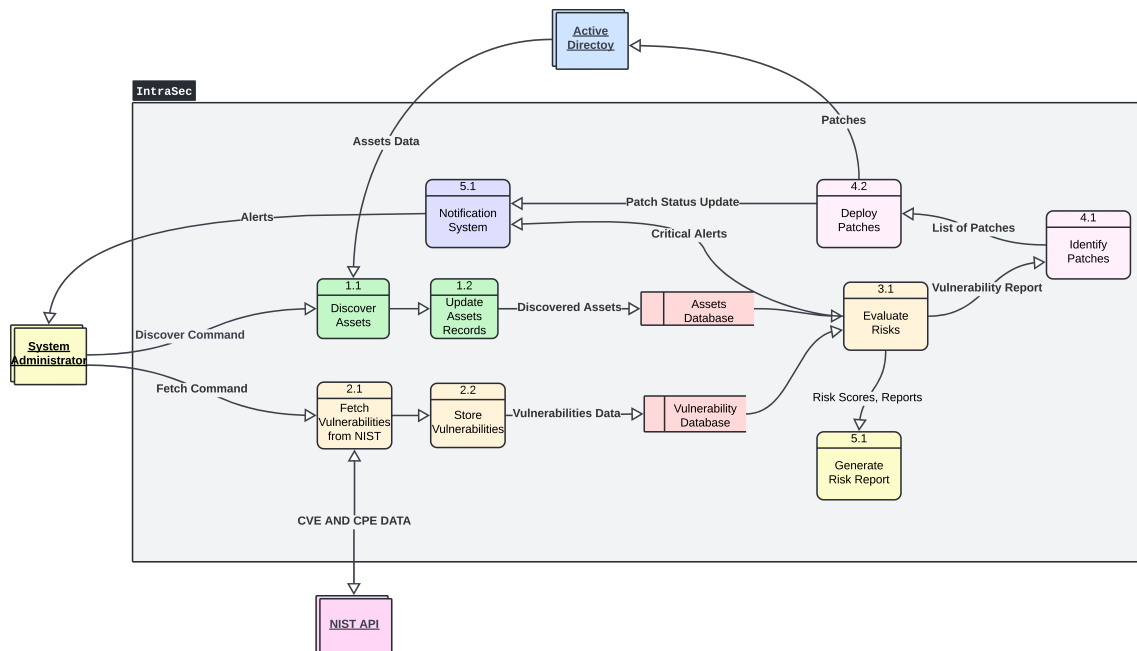


Figure 3.4: Data-flow Diagram - Level 2

3.2.2 Activity Diagrams

3.2.2.1 Discover Assets

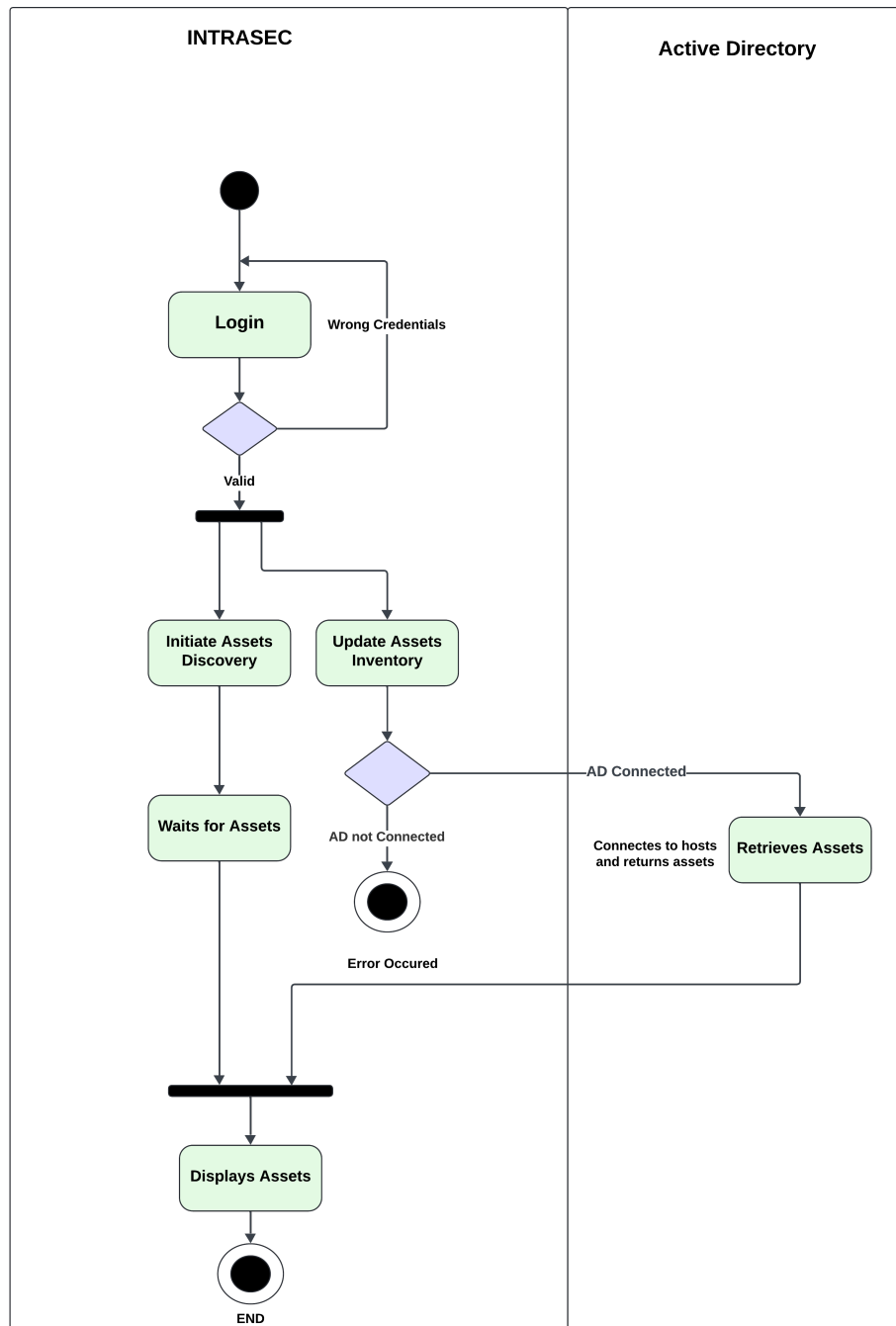


Figure 3.5: Discover Assets - Activity Diagram

3.2.2.2 Assess Risks

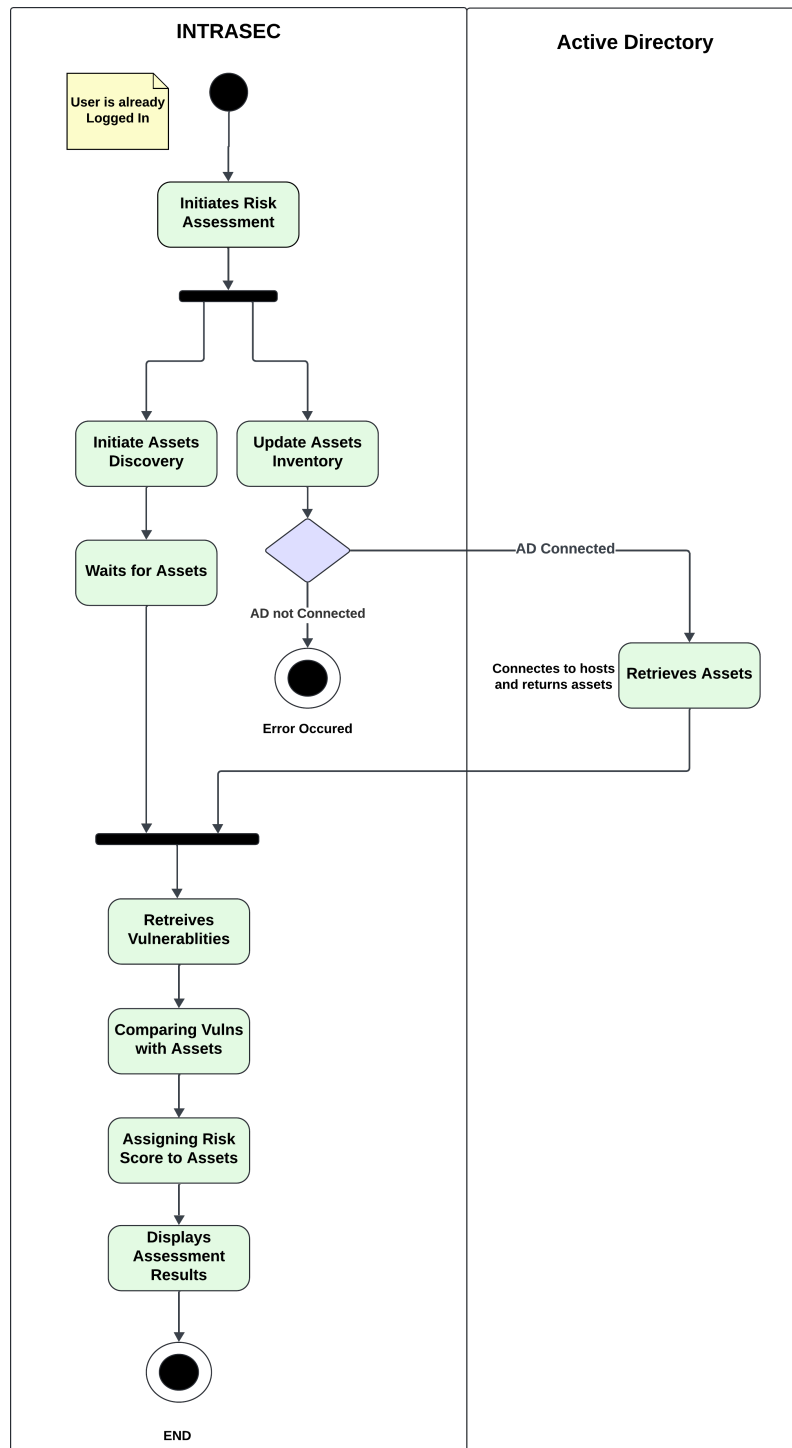


Figure 3.6: Assess Risk - Activity Diagram

3.2.2.3 Generate Report

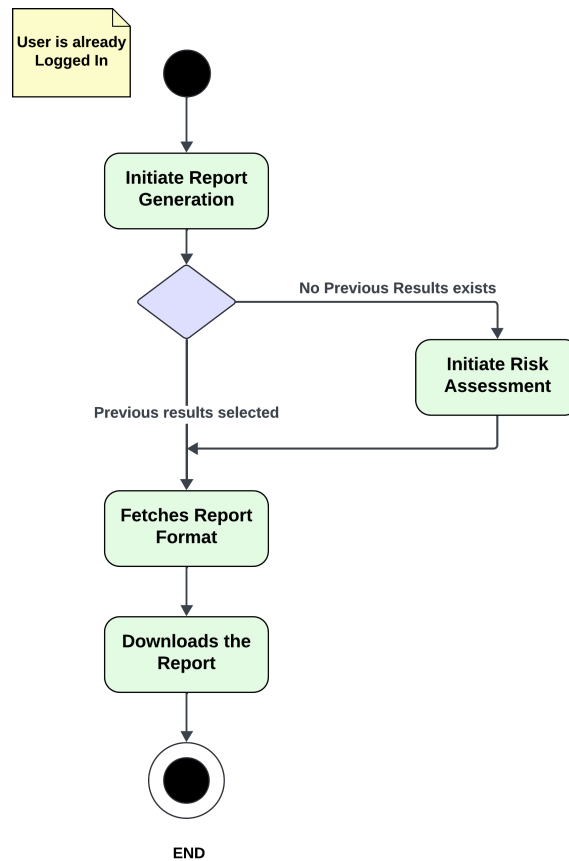


Figure 3.7: Generate Report - Activity Diagram

3.2.2.4 Deploy Patches

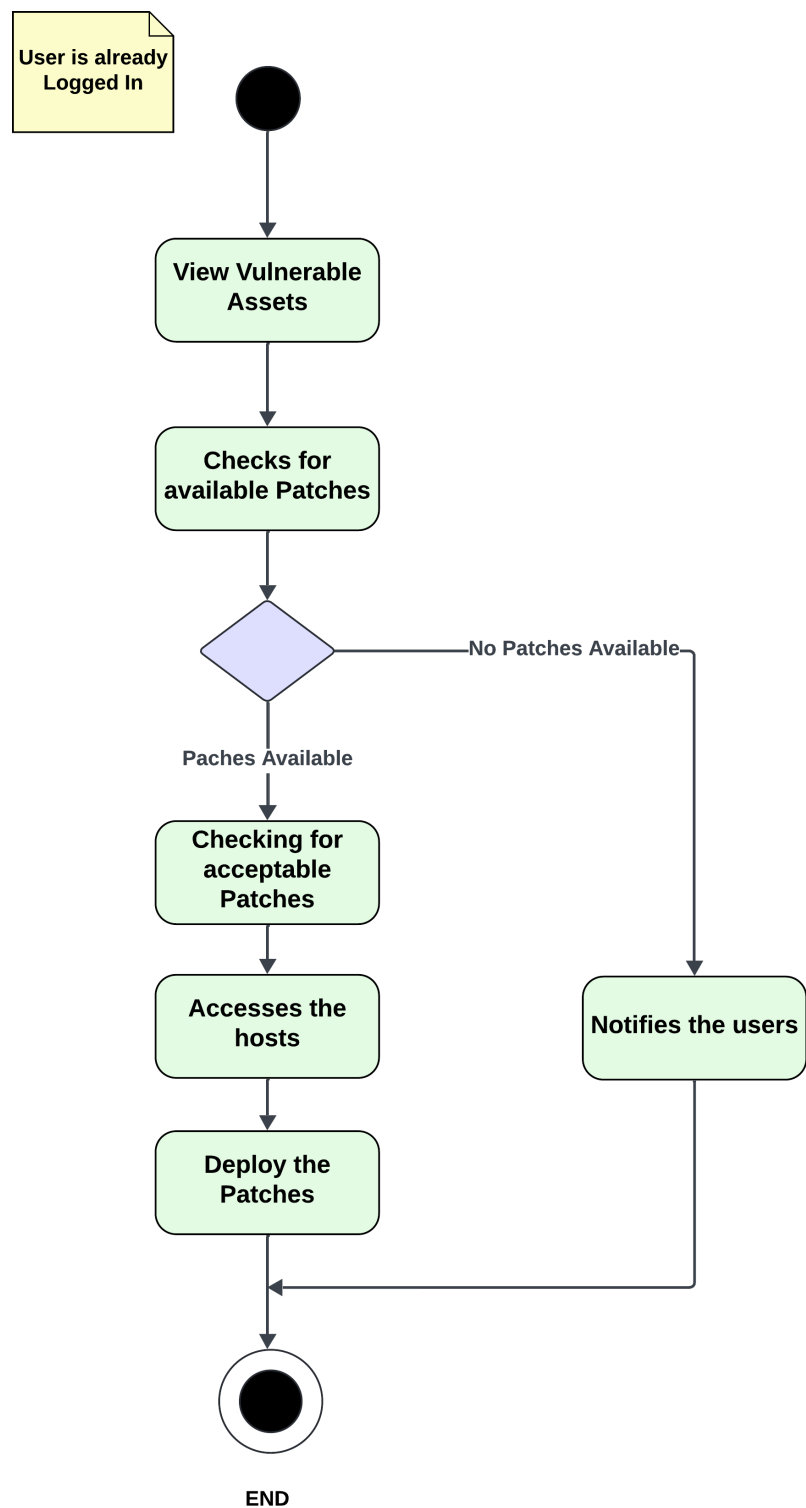


Figure 3.8: Deploy Patches - Activity Diagram

3.2.3 System-level Sequence Diagrams

3.2.3.1 Discover Assets

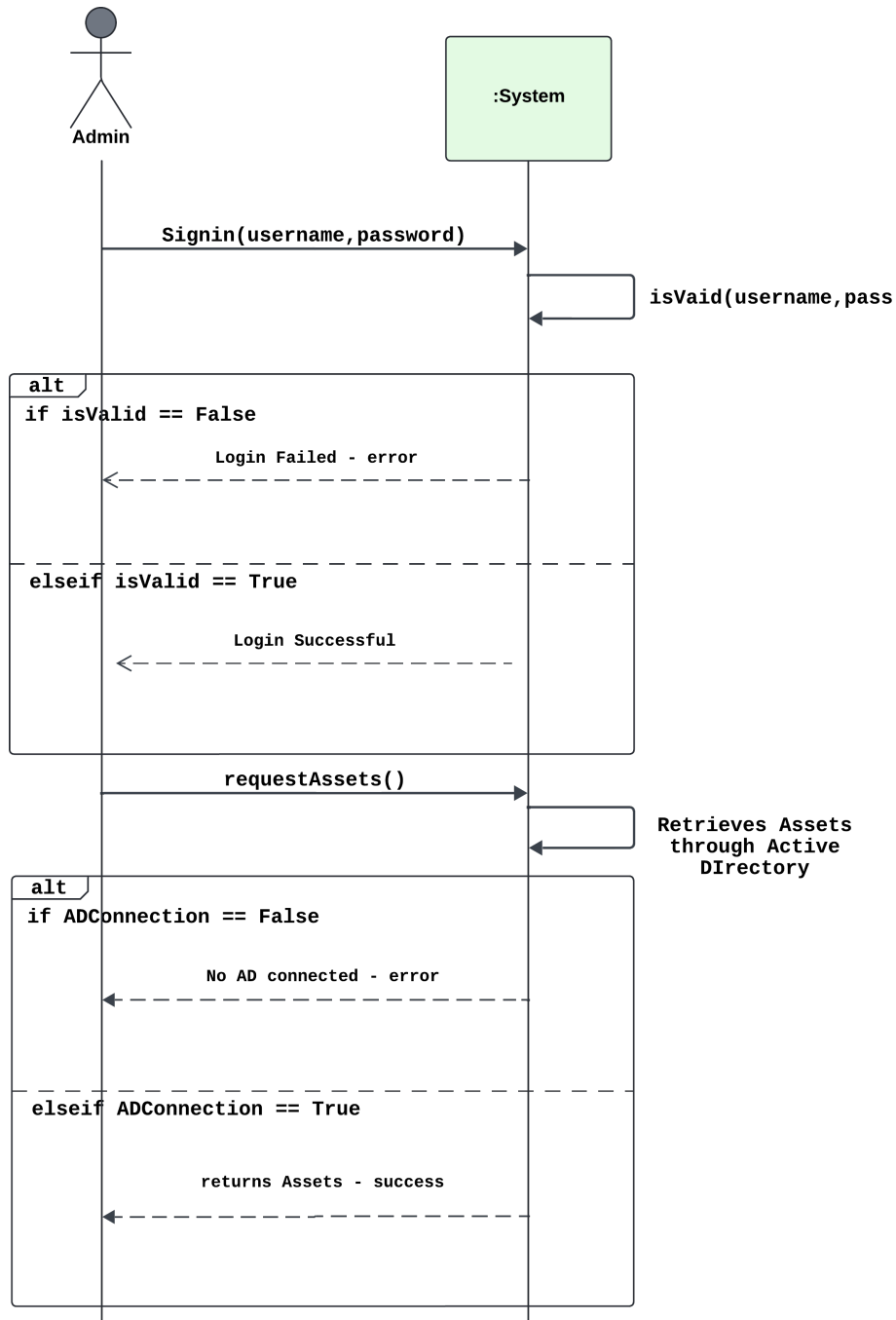


Figure 3.9: Discover Asset - System Sequence Diagram

3.2.3.2 Manage Assets

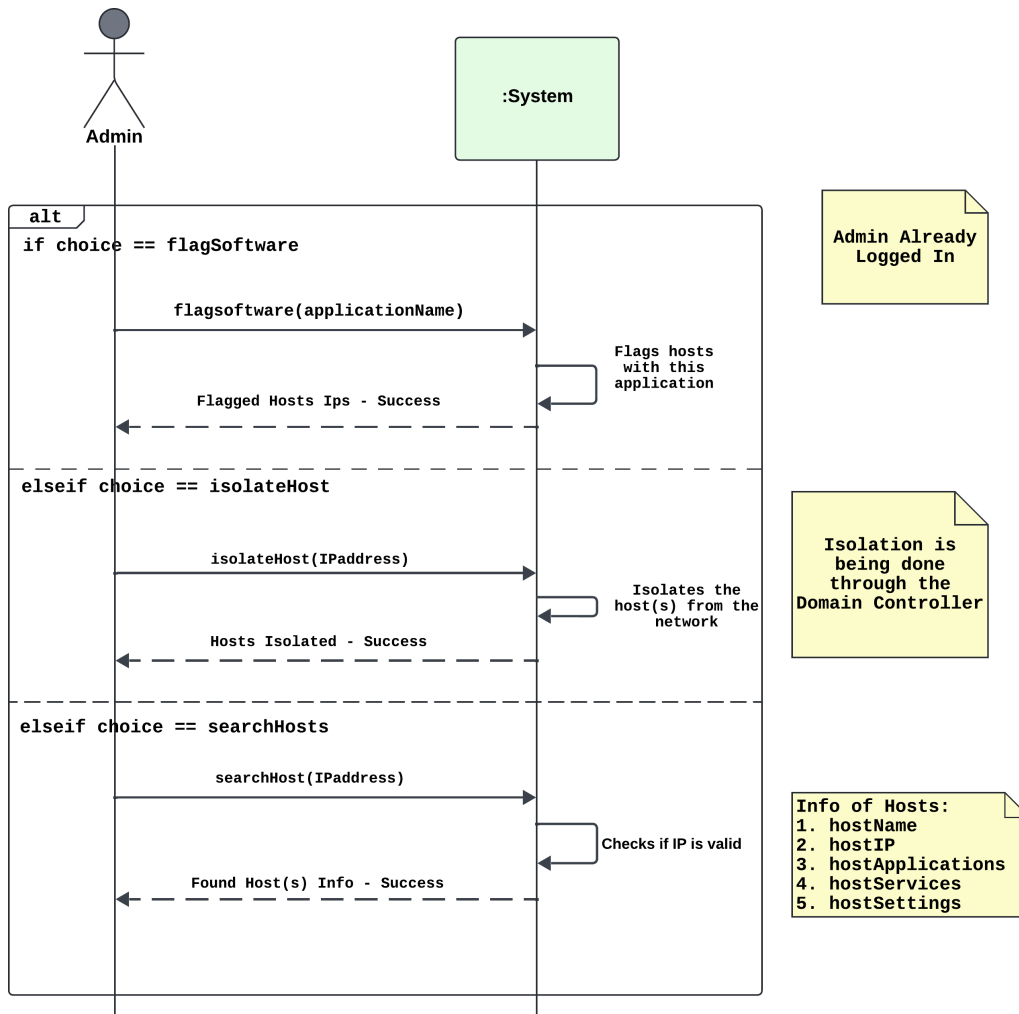


Figure 3.10: Manage Assets - System Sequence Diagram

3.2.3.3 Deploy Patches

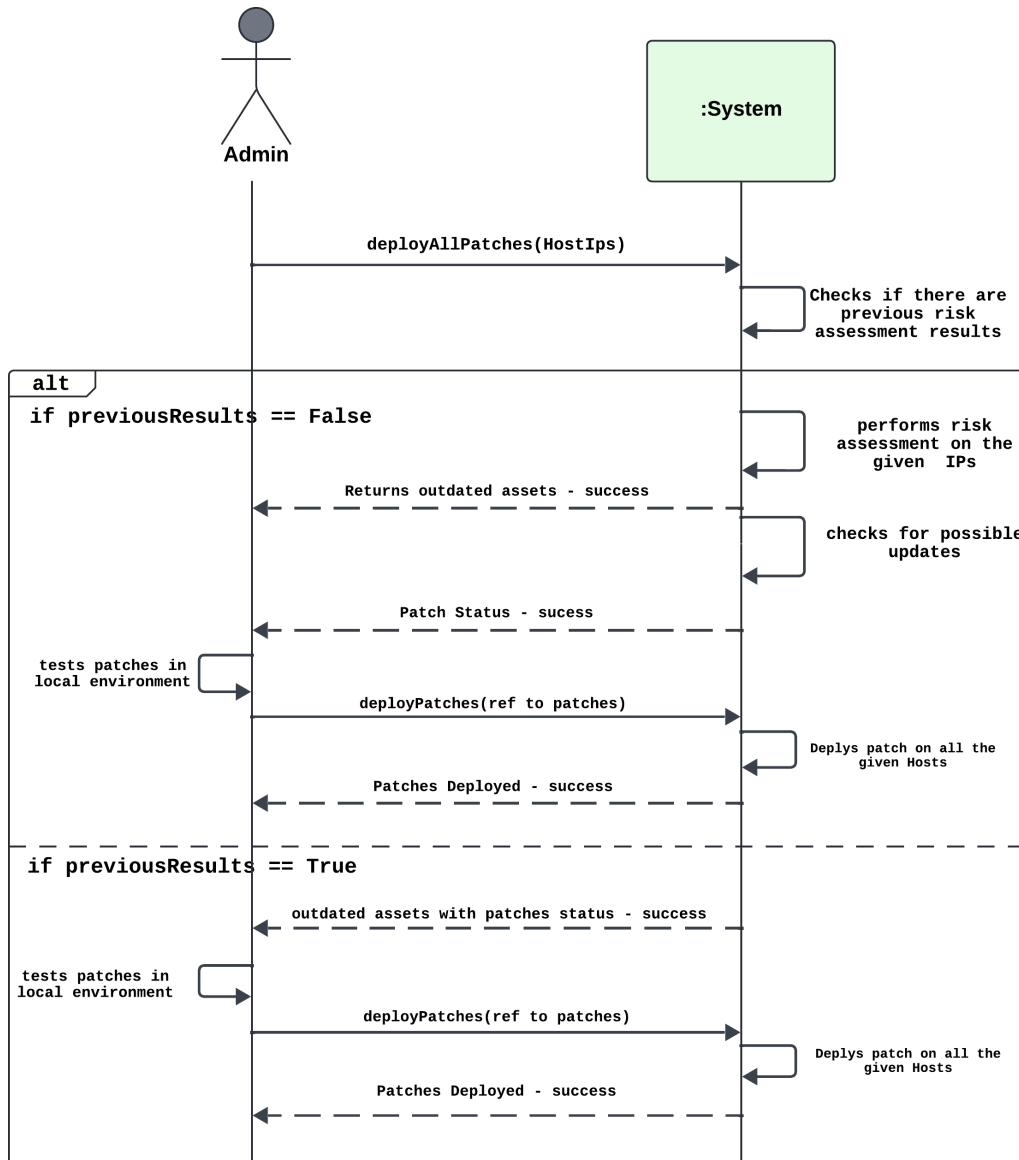


Figure 3.11: Deploy Patches - System Sequence Diagram

3.2.3.4 Assess Risks

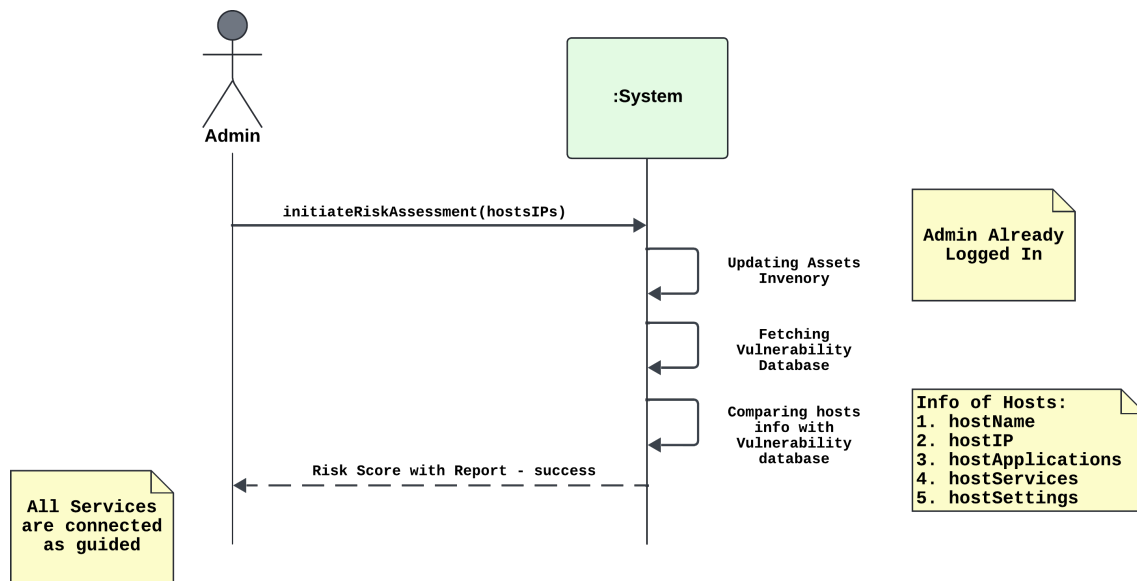


Figure 3.12: Assess Risks - System Sequence Diagram

3.3 Data Design

The information domain in our system consists of vulnerabilities represented as **Common Vulnerabilities and Exposures (CVEs)** and **Common Platform Enumerations (CPEs)**, along with related entities such as assets and patches. This data is subject to constant updates.

3.3.1 Data Structures

In our system, the main entities and their relationships can be represented as follows:

3.3.1.1 Key Entities

- **CVEs:** Common Vulnerabilities and Exposures, which represent identified vulnerabilities.
- **CPEs:** Common Platform Enumerations, which represent specific hardware and software configurations affected by vulnerabilities.
- **Assets:** Represent devices in the network that may be affected by vulnerabilities.
- **Patches:** Represent fixes or updates for the vulnerabilities.
- **Reports:** Reports based in risk assessment results

3.3.2 Elasticsearch Indexes

In Elasticsearch, data is organized in the form of indexes. The structure might include:

- **Vulnerabilities Index:**

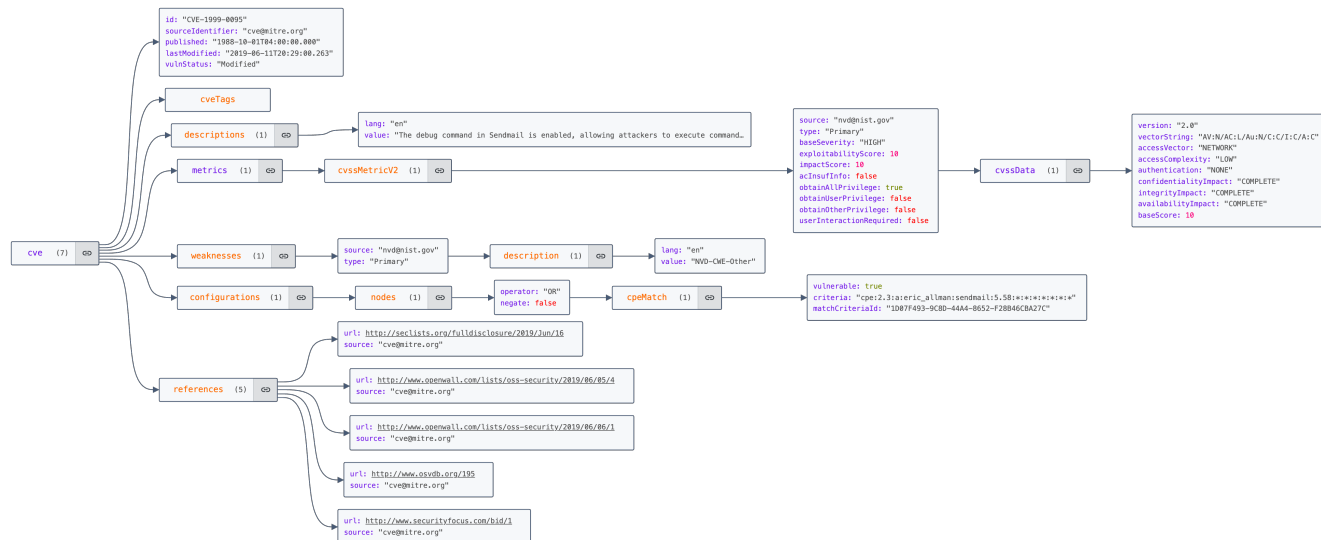


Figure 3.13: CVE JSON Schema

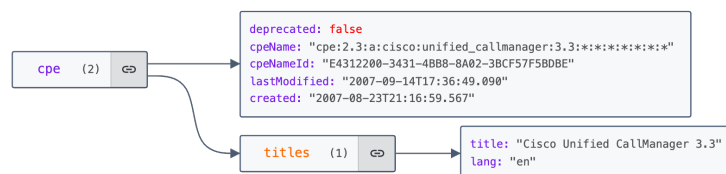


Figure 3.14: CPE JSON Schema

– Document Type: cve

- * **cve_id** (string): Unique identifier for the CVE.
- * **description** (text): Description of the vulnerability.
- * **severity** (keyword): Severity level of the vulnerability (e.g., low, medium, high).
- * **published_date** (date): Date when the CVE was published.
- * **cpe_list** (nested): Array of CPEs associated with the CVE.

3.3.3 PostgreSQL Databases:

In our system, we utilize PostgreSQL to store data in structured tables. The structure might include:

- **Assets DB:**
 - Few of the attributes of this table are mentioned below:
 - * `asset_id` (string): Unique identifier for the asset.
 - * `asset_type` (keyword): Type of asset (e.g., server, workstation).
 - * `status` (keyword): Status of the asset (e.g., active, inactive).
 - * `cve_associated` (array): Array of CVEs associated with the asset.
- **Patches DB:**
 - Few of the attributes of this table are mentioned below:
 - * `patch_id` (string): Unique identifier for the patch.
 - * `cve_id` (string): CVE that the patch addresses.
 - * `release_date` (date): Date when the patch was released.
- **Reports DB:**
 - Few of the attributes of this table are mentioned below:
 - * `report_id` (string): Unique identifier for the report.
 - * `report_date` (date): Date when the report was generated.
 - * `asset_id` (string): Unique identifier of the asset related to the report.
 - * `cve_id` (string): CVE identifier associated with the vulnerabilities addressed in the report.
 - * `summary` (text): Brief summary of the findings in the report.
 - * `recommendations` (text): Suggested actions or recommendations based on the report's findings.

Above is the information domain as per iteration 1 with current data requirements. Minor updates might be made when work starts on specific modules in the future.

Bibliography